Simulation of Classroom Speech by Elementary School Teachers Using a Large Language Model with LoRA Hyperparameter

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***Abstract*—Reflective activities are considered important for teachers to improve their teaching. However, it is difficult to evaluate reflection activities objectively because it is necessary to consider self-reflective aspects. The goal of this research is to systemize a simulation in which the teacher's speech is simulated by other teachers to provide feedback to the teacher. It is also increasingly becoming possible to generate speech using large language models. Therefore, in order to simulate the speech of other teachers, a large language model based on LoRA fine-tuning for each teacher was used, and the speech in the class was generated by the language model of another teacher.**

***Index Terms*—Classroom Utterance Analysis, Dialogue System, Large Language Model (LLM), Low Rank Adaptation (LoRA), Utterance Simulation**

I. Introduction

In terms of class improvement by teachers, training programs such as research classes and reflective activities [1]–[3] have been conducted. While reflective activities have attracted attention in particular, it is difficult to evaluate them objectively because of the need to consider self-reflective aspects in reflection activities. Therefore, there are calls for a system to assist teachers' reflection for which objective evaluation is possible. The number of studies that support teachers is smaller than those that support learners, and even smaller if we limit these to studies that assist teachers in their reflections. In the field of natural language processing, the Large Language Model (LLM), among other types of sentence generation, is used for facilitating speech generation. In addition, whereas tuning of large language models requires a GPU with a large amount of VRAM, Low Rank Adaptation (LoRA) [4] enables tuning with a reduced amount of VRAM memory. In contrast to this situation, in actual elementary school classes, the teachers observe the children's situation and proceed with the class, and the children often talk about their learning situation, opinions, and impressions. This means that the class will proceed through dialogue between the teacher and the children. Therefore, by simulating what other teachers would produce in a certain classroom dialogue, it would be possible to analyze and give advice on speech production. The goal of this study is to develop a simulation system that generates highly proficient teacher utterances in response to teacher utterances in class. In order to simulate the speech of other teachers, we collected the speech of teachers and children in class and simulated the generation of speech by another teacher's model using a large language model for each teacher that was fine-tuned by LoRA. In particular, we are training LoRA using prompts and simultaneously applying LoRA trained on all teachers' utterances and LoRA trained on individual teachers. LoRA hyperparameter optimization is also performed by changing the type of weights to which LoRA is applied and the range of layers to which LoRA is applied as LoRA hyperparameters.

II. Related Research

Although there is research on teacher support that analyzes teacher speech in terms of linguistic and nonlinguistic features [5], this is less common than research that supports learners, such as research on intelligent learning support systems that apply dialogue systems to education [6].

In particular, there have been few studies to assist teachers' reflection, and machine learning research [7] and speech analysis systems that recognize students' lecture situations [8] have only just started to be proposed. With regard to elementary school classes, there is a study [9] that analyzed the digitalization of reflection methods from the perspective of "independent, interactive, and deep learning".

In the field of natural language processing, the Global Variational Transformer (GVT) [10], which introduces a transformer [11] to Conditional Variational AutoEncoder (CVAE) [12] for dialogue, has been proposed. We propose an extended GVTSC model [13] that adds latent variables for each speaker and abstraction of speaker features by clustering in order to take speaker characteristics into account. In contrast to this, Llama 2 [14] and other LLM-based conversational response generators have also been proposed.

III. Overview of Dialogue Data

The dialogue data of a 45-minute math class at an elementary school was recorded, and the teacher and children's speech

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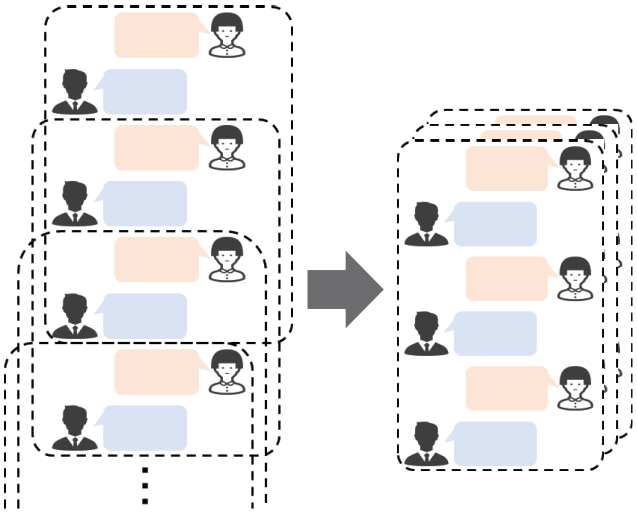


Fig. 1. Dataset preprocessing

was transcribed to create textual information in dialogue format. This study covers three types of lessons, the subject of each of which is math. Teacher A' s lesson was a 4th grade class looking at how to find the length around a staircase. Teacher B' s lesson was a 6th grade class on finding volume relationships. Teacher C' s lesson was a 5th grade class on how to divide decimals with remainders.

Preprocessing of dialogue data is shown in Figure 1. Figure 1 For each class, dialogue data consisting of one dialogue between the teacher and a child for the entire class is created. The entire class dialogue was divided into a short dialogue in which the sixth and final utterance was the teacher's utterance, sliding by two utterances within the range of six utterances, and a dataset was created for generating teacher dialogue responses. In this study, four datasets were used: three dialogue datasets for each teacher and a merged dataset of all teachers' dialogues. The number of data in the four datasets was 96 for Teacher A, 152 for Teacher B, 72 for Teacher C, and 320 for all teachers combined. Each dataset was subjected to the Train:Validation:Test at a ratio of 8:1:1 for fine-tuning at a ratio of 8:1:1 by LoRA.

IV. Overview of LLM and LoRA used

Currently, a number of LLM models have been developed, and Youri 7B, which was continuously pre-trained on Japanese data, is available for Llama 2 7B [14]. In this study, fine tuning of LLM using LoRA is conducted, and Youri 7B Chat [15], which was additionally trained on multi-turn dialogue data, is used as the base model to handle Japanese dialogue. In LLM, fine tuning of all parameters requires extensive computational resources. LoRA updates only small parameters added to the LLM, allowing for fine-tuning with less computational resources and time. QLoRA [16], which requires less computing resources and time, trains LoRA with 4-bit quantization of the base model.

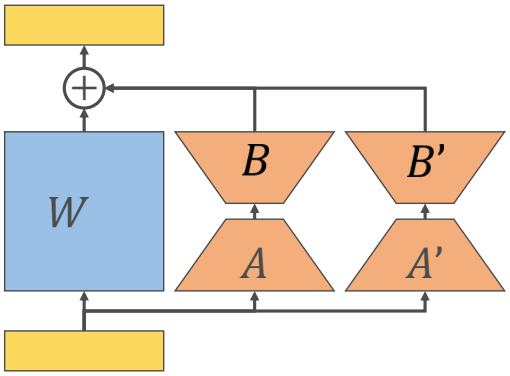


Fig. 2. Structure during inference when multiple LoRA are applied

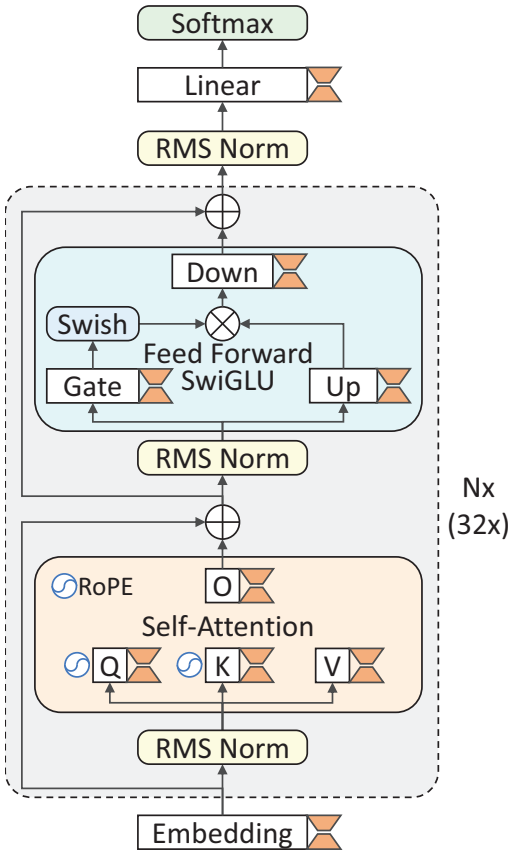


Fig. 3. Structure of Llama 2 and applicable weights of LoRA

Figure 2 shows the structure of inference when multiple LoRAs are applied. In addition to the LLM weights W, the LoRA1 weights A and B and the LoRA2 weights A′ and B′ are used for inference. Also, the structure of Llama 2 and applicable weights of LoRA are shown in Figure 3. The applicable weights of LoRA are represented by square elements with a white background color, and are indicated by adding a trapezoidal weight representing LoRA next to it. Llama 2 7B has a structure with 32 layers of gray background

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TABLE I
HYPERPARAMETER EXPERIMENTAL SETUP

Applied weight	
2	W_q, W_v
7	$W_q, W_k, W_v, W_o, W_{up}, W_{gate}, W_{down}$
9	$W_q, W_k, W_v, W_o, W_{up}, W_{gate}, W_{down}, W_{emb}, W_{vocab}$
Applicable layer	
1–32	1–8, 9–16, 17–24, 25–32
9–32	9–16, 17–24, 25–32
17–32	17–24, 25–32
25–32	25–32

color areas surrounded by dotted lines.

V. LoRA HYPERPARAMETER OPTIMIZATION

A. Hyperparameter Experimental Setup

In this study, the hyperparameters of LoRA in the dialogue response generation task are optimized using the dialogue dataset of all teachers before simulating teachers' speech. The LoRA hyperparameters are **(1)** the type of weights to which LoRA is applied and **(2)** the range of layers to which LoRA is applied. The other hyperparameters, namely the low-rank matrices r and scaling α, are fixed at 8 and 16, respectively. Table I shows the hyperparameters for which combinations were tested in this experiment.

(1) LoRA applied weights As shown in Figure 3, LoRA can be applied to the 9 types of elements represented by the squares. The types of weights to which LoRA can be applied are Self-Attention's query (Q), key (K), value (V), output (O) weight W_q, W_k, W_v, W_o, Feed Forward's up, gate, down weight W_{up}, W_{gate}, W_{down}, Embedding weights W_{emb} and vocabulary output layer (Linear) weights W_{vocab}. In this experiment, in addition to the 2 types of weights in the original LoRA [4] setting, we have tried combinations of 7 types, excluding the weights responsible for the conversion to language, and all 9 types, including the weights responsible for the conversion to language. The 2, 7, and 9 types of weights, respectively, are listed in Table I.

(2) LoRA applied layers LoRA allows the user to specify the layers to be fine-tuned in addition to the type of weights. Since Llama 2 7B, used in this experiment, has a 32-layer structure with N in Figure 3, the range of applicable LoRA layers was changed by dividing the 32 layers into four equal parts, eight layers at a time, and an experiment was conducted to evaluate the impact on the task of generating dialogue responses. As shown in Table I, we tried a combination of settings that apply LoRA to all layers from 1 to 32 (1–32), layers 9 to 32 (9–32), layers 17 to 32 (17–32), and layers 25 to 32 (25–32). In other words, 1–32 applies LoRA to all layers, 9–32 to the back 24 layers, 17–32 to the back half of 16 layers, and 25–32 to the back 8 layers only.

Experiments were conducted with a total of 12 hyperparameter configurations, which consisted of 3 combinations of LoRA applied weights and 4 combinations of LoRA applied layers. Note that Embedding W_{emb} and Linear W_{vocab} are outside the range of 32 applied layers, as shown in Figure 3.

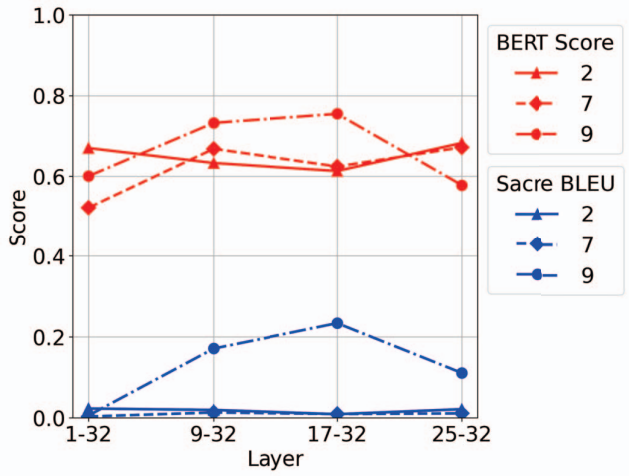


Fig. 4. Impact of Hyperparameters on Evaluation

B. Hyperparameter Optimization Evaluation Experiment

In the evaluation experiment, fine tuning was performed by LoRA using the dialogue dataset of all teachers, and the performance of dialogue response generation was evaluated by BERT Score [17] and Sacre BLEU [18], [19]. The BERT Score is an automatic evaluation index that uses BERT to evaluate the similarity between a generated response and a reference response. Sacre BLEU is an automatic evaluation index that evaluates the similarity between a generated response and a reference response using N-gram agreement.

Evaluation results are shown in Table II. In Table II, the BERT Score and Sacre BLEU are shown in parentheses. The settings in which LoRA was applied to all weights in the latter 16 layers resulted in the highest BERT Score and Sacre BLEU of 0.7549 and 0.2338, respectively, for both evaluations. The difference between the lowest and highest BERT Score was 0.2329, suggesting the importance of hyperparameter search in LoRA.

Figure 4 shows a line plot of the applied weights, with BERT Score and BLEU on the vertical axis and the applied layers on the horizontal axis. A similar trend is observed for BERT Score and BLEU. Table III shows the variation in the number of LoRA parameters. The highest number of parameters that can be learned is 1–32 for all layers, but this is not a high evaluation by any means. The number of parameters for LoRA applied to 9 types of weights in the 1–32 layer has a maximum of 21M and a minimum of 1M in the 25–32 layer with 2 types of weights setting. The BERT Score is 0.6004 and 0.6827, respectively, resulting in a 0.08 lower rating despite the 21–maximum number of parameters. When the applied weight is 9, the parameters of the Embedding layer before layer 1 and the output layer of the vocabulary after layer 32 are also learned, but the model that learned 32 is rated higher. This suggests that the effect of learning the parameters of the Embedding layer could be handled by learning only the parameters from layer 17 onward.

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TABLE II
EVALUATION OF HYPERPARAMETER OPTIMIZATION

Applied weight	Applicable layer 1–32	9–32	17–32	25–32
2	0.6700 (0.0230)	0.6329 (0.0195)	0.6130 (0.0090)	0.6827 (0.0214)
7	0.5220 (0.0033)	0.6681 (0.0133)	0.6242 (0.0094)	0.6719 (0.0111)
9	0.6004 (0.0058)	0.7324 (0.1711)	**0.7549** (**0.2338**)	0.5780 (0.1108)

TABLE III
VARIATIONS IN THE NUMBER OF LORA PARAMETERS

Applied weight	Applicable layer 1–32	9–32	17–32	25–32
2	4,194,304	3,145,728	2,097,152	1,048,576
7	19,988,480	14,991,360	9,994,240	4,997,120
9	20,566,016	15,568,896	10,571,776	5,574,656

TABLE IV
RESULTS OF EVALUATION OF DIALOGUE RESPONSE GENERATION BY THE SAME TEACHER

Model	Data	BERT	BLEU	Appropriateness
LoRA 1	Teacher A	0.7397	0.0326	3.22
LoRA 2	Teacher B	0.6303	0.1017	2.89
LoRA 3	Teacher C	0.7946	0.1186	**4.00**
LoRA 4	All teachers	0.7549	0.2338	2.56
LoRA 5	Teacher A with P	**0.7948**	0.2212	3.67
LoRA 6	Teacher B with P	0.7774	**0.4887**	3.11
LoRA 7	Teacher C with P	0.7081	0.1039	2.78
LoRA 8	All teachers with P	0.7174	0.2770	2.78
Base	Teacher A	0.6096	0.0000	-
Base	Teacher B	0.1163	0.0000	-
Base	Teacher C	0.5100	0.0000	-
Base	All teachers	0.3484	0.0000	-
-	Actual teacher	-	-	4.25

VI. IMPACT OF PROMPTS

In LLMs, it is understood that performance can be improved by the designing of prompts [20]. We are experimenting with the improvement of the performance of LLM fine-tuning using LoRA by restricting domains with prompts. The prompt is "You are Teacher A of an elementary school." and the keywords "elementary school" and "teacher" are used in Japanese to limit the domain.

Furthermore, in the training using the dialogue dataset of all teachers, the dialogue of multiple teachers is trained into a single model. In such a case, by changing the prompt for Teacher A to A, B, and C for each teacher, we expect that using the prompt for Teacher B will generate a response as Teacher B. In other words, the possibility of switching between teachers by changing the prompts in the generation is evaluated.

VII. TEACHER SPEECH SIMULATION

A. Training of teachers' dialogue response generation models

LLM LoRA fine-tuning was conducted using four datasets – one for each teacher and one for all teachers. For the LoRA hyperparameters, we use the setting where LoRA is applied to all weights in the latter 16 layers, which showed the best performance in BERT Score and Sacre BLEU in Section V. In addition, when fine tuning, we have also created a model that improves the performance of dialogue response generation by restricting the domain described in section VI with prompts.

Table IV shows the results of evaluating the performance of dialogue response generation within the same teacher. Models can be broadly divided into three categories. LoRA 1–4 are models trained on LoRA using Data. LoRA 5–8 is a model of LoRA trained with Data and prompts. The "with P" in the Data in Table IV indicates that a prompt was used. Base is the result of evaluating the base model as a baseline without any fine tuning. The BERT Score and Sacre BLEU are used as automatic evaluation indices, and manual evaluation is also performed. In the manual evaluation, three people rated the appropriateness of the response to the context on an absolute scale of 1–6, with a rating of 6 being the most appropriate. In manual evaluation, 3 dialogues are sampled from the Test dataset and evaluated. The evaluation with LoRA was confirmed to improve BERT Score and BLEU compared to the evaluation without LoRA, suggesting the effectiveness of fine tuning with LoRA.

(1) LoRA 1–4 Except for the model trained on the dialogue dataset of Teacher B, all the other models obtained a BERT Score of 0.7 or higher. LoRA 4, which merges all teachers, received a high rating of 0.7549, despite the need to infer the responses of different classes and teachers, suggesting that the semantic similarity according to BERT was evaluated as having many parts in common among classes and teachers. In the appropriateness evaluation, Teacher C from LoRA 3 was rated prominently higher, similar to the BERT Score, which is close to the actual teacher's evaluation.

(2) LoRA 5–8 with P LoRA 5 and LoRA 6 received the highest BERT Score and BLEU, respectively, with Lora 6 's BLEU being particularly high, while Lora 6, with its higher BERT Score, received a higher rating for appropriateness. Unlike the model in LoRA 1–4, all teachers received lower ratings, but this may have been due to confusion caused by changing the prompts for each teacher. The result for the appropriateness of the response also shows a low rating of 2.78, which is similar to the BERT Score.

(3) Effectiveness of prompts (with P) The BERT Score

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TABLE V
THE AVERAGE RATING OF APPROPRIATENESS BY MANPOWER

Model	Appropriateness
Each teacher	2.63
All teachers + each teacher	1.72
Each teacher with P	2.69
All teachers with each teacher P	**2.83**
Actual teacher	4.11

improved for Teacher A and Teacher B, and the BLEU improved for all except Teacher C, suggesting the effectiveness of the prompts. In particular, Teacher B improved BERT valuation index was observed in the evaluation of appropriateness by manpower. By setting the domain as "elementary school teachers" in the prompt, the broad domain of the LLM can be limited to the scope related to Japanese elementary school teachers, and fine-tuning is considered to have been effective.

B. Evaluation Methods for Teacher Speech Simulation

In this study, we simulate how a teacher that differs from the actual teacher would respond during actual classroom dialogue in that situation. In the teacher's speech simulation, the eight models described in Section VII-A are used to generate response utterances to the context, which are then evaluated manually. The data to be evaluated consisted of six conversations, two from each of the three teachers' test datasets. In the manual evaluation, the appropriateness of the response utterances to the context was rated by three participants on an absolute scale of 1–6, with 6 being the most appropriate response.

There are 12 models for generating responses: three models for each teacher, three models for all teachers and three models for each teacher applied simultaneously, three models for each teacher using prompts, and three models for all teachers using prompts for each teacher. In the case of all teachers + each teacher, as shown in Figure 2, the LoRA of all teachers and the LoRA of each teacher are applied simultaneously to test whether it is possible to use the knowledge, etc., of all teachers in addition to each teacher. For all teachers with each teacher P, when simulating Teacher A, the prompt is "You are elementary school Teacher A." and expects all teachers to behave as Teacher A and have the knowledge of all the teachers, etc.

C. Evaluating the Appropriateness of the Human Response

Table V shows the average rating of appropriateness by manpower. In order to evaluate differences in methods rather than differences among teachers, the average rating for each method is shown. All teachers and each teacher who applied LoRA to all teachers+each teacher at the same time often had repetition, and the manpower evaluation of appropriateness was also low at 1.72. All teachers with each teacher P received the highest rating of 2.83 in the simulation of teacher speech, although the evaluation in Section VII-A suggested the possibility of confusion due to differences among teachers, and that the response format was sometimes incorrect. This suggests that learning from the dialogue data of all the teachers may have increased the versatility of the system. The appropriateness of the response uttered by the actual teacher was 4.11, a large difference considering that the appropriateness of the response generated by the simulation was less than 3.

TABLE VI
EXAMPLES OF TEACHER SPEECH SIMULATIONS

Context
Student: ありえない。 (It's Impossible.) **Teacher C:** ありえ? (Imposs?) **Student:** ない! (ible!) **Teacher C:** そこら辺も含めて、じゃあ、ここはあまり 0.3M で? (So, including that, this is the left over 0.3M?) **Student:** いい。あってる。 (It's fine. It fits.)
Response
Teacher A: 合ってる? (Am I right?) **Teacher B:** システムな? (System?) **Teacher C:** 合ってそう?ほんでな、 (Does it seem to fit? What, really?) **All teachers + teacher A:** .2? (.2?) **All teachers + teacher B:** システム:0.3M あこのこのこの

VIII. Conclusion

In this study, with the goal of providing feedback to teachers, we conducted a simulation in which teachers' LLMs fine-tuned by LoRA were used to generate utterances in a certain situation as another teacher. The model, trained with all teachers' dialogue data with the domain restricted by the prompt, received a high evaluation, possibly due to its high versatility. For the LoRA hyperparameter optimization, the setting in which LoRA is applied to all weights in the latter 16 layers received the highest rating.

Future issues include improving the lack of information due to the small number of utterances in the context, collection and construction of training data, tuning of prompts, and methods of evaluating teachers' speech simulation.

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